

SASTRA DEEMED UNIVERSITY
TIRUMALAISAMUDRAM, THANJAVUR – 613 401



Manual for Engineering Drawing
Module 1

Name:

Register Number:



SASTRA

ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION

DEEMED TO BE UNIVERSITY

(U/S 3 of the UGC Act, 1956)



THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

T H A N J A V U R | K U M B A K O N A M | C H E N N A I



BEST
BAJAJ
ENGINEERING
SKILLS
TRAINING

BONAFIDE CERTIFICATE

Certified that this is the bonafide record of work done by Chiranjeevi/Sowbagyavathi

with Register No. _____ who studied/practised in
the _____ Laboratory during
the year _____

Staff in Charge

Assessment Pattern

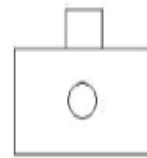
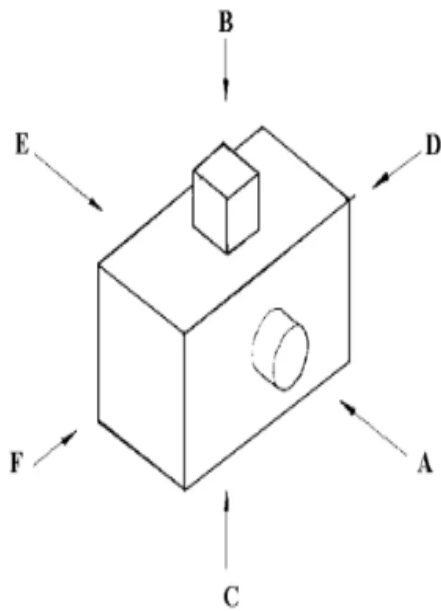
Assessment Mark Split up	Marks	Weightage
<u>Internal Assessment</u> Drawing/Exercise practice during lab hours using ARES Commander Software	20	70 (LAB)
<u>External Assessment</u> 1.5 Hours - Drawing – 40 marks - Tools & Procedure followed – 10 marks.	50	
<u>External Assessment</u> 30 minutes MCQs/Short answers – 30 marks	30	30 (THEORY)
Total		100

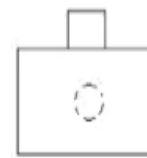
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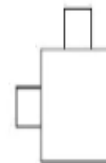
Sl.No	Description	Date	Marks	Sign
1	Basic Understanding of Various Orthographic Views			
2	Isometric to Orthographic Views – First angle projection			
3	Isometric to Orthographic Views – Third angle projection			
4	Industrial Drawing with GD & T and Cross-sections			
5	3D Model Exercises			

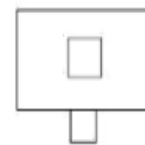
Exercise Number 1: Basic Understanding of Various Orthographic Views

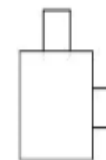
EXERCISES. In which direction must the object be viewed to produce the views shown opposite, taking 'A' as the FRONT VIEW. Put the appropriate letter under the view.

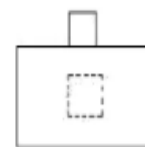










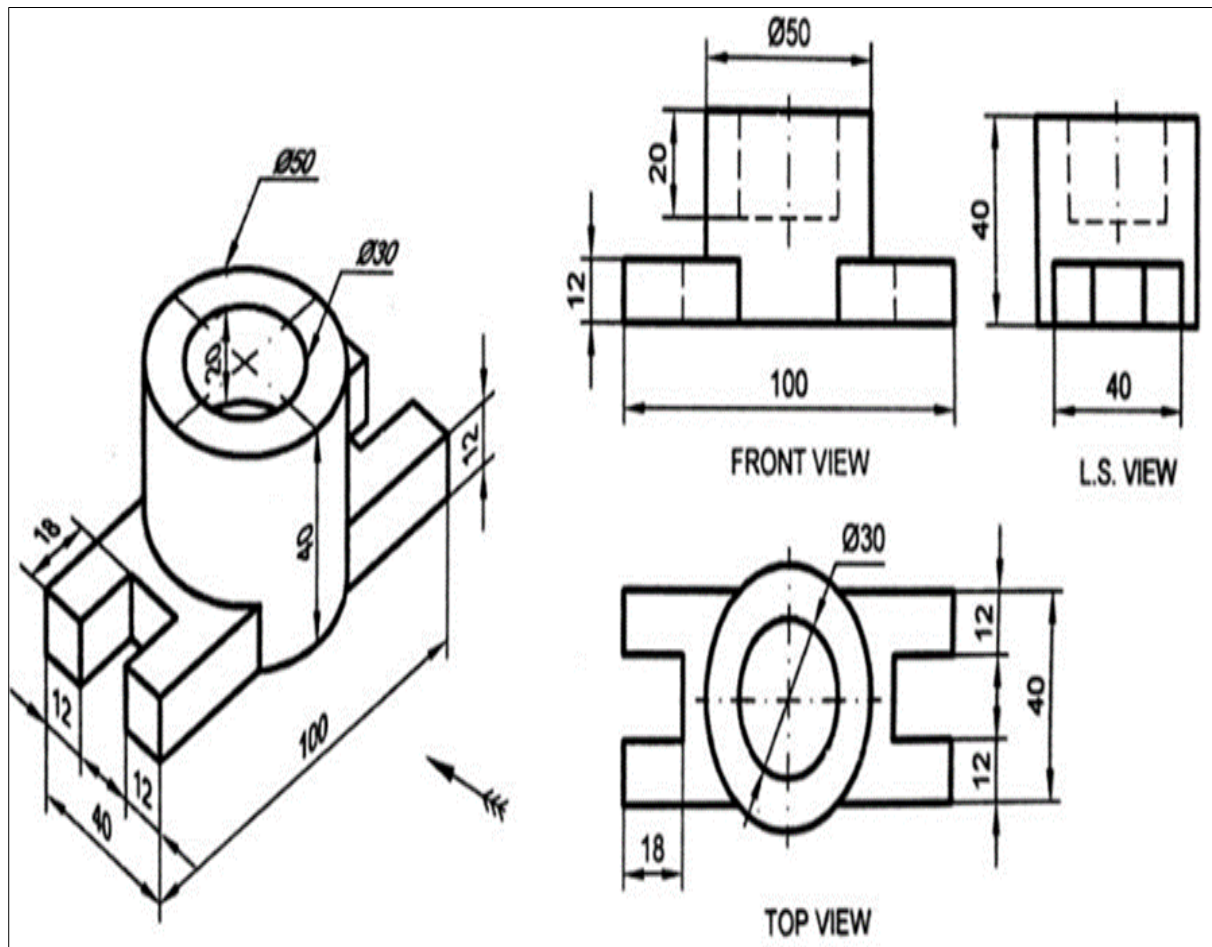


DRG.

ORTHOGRAPHIC PROJECTION

EXERCISE 1

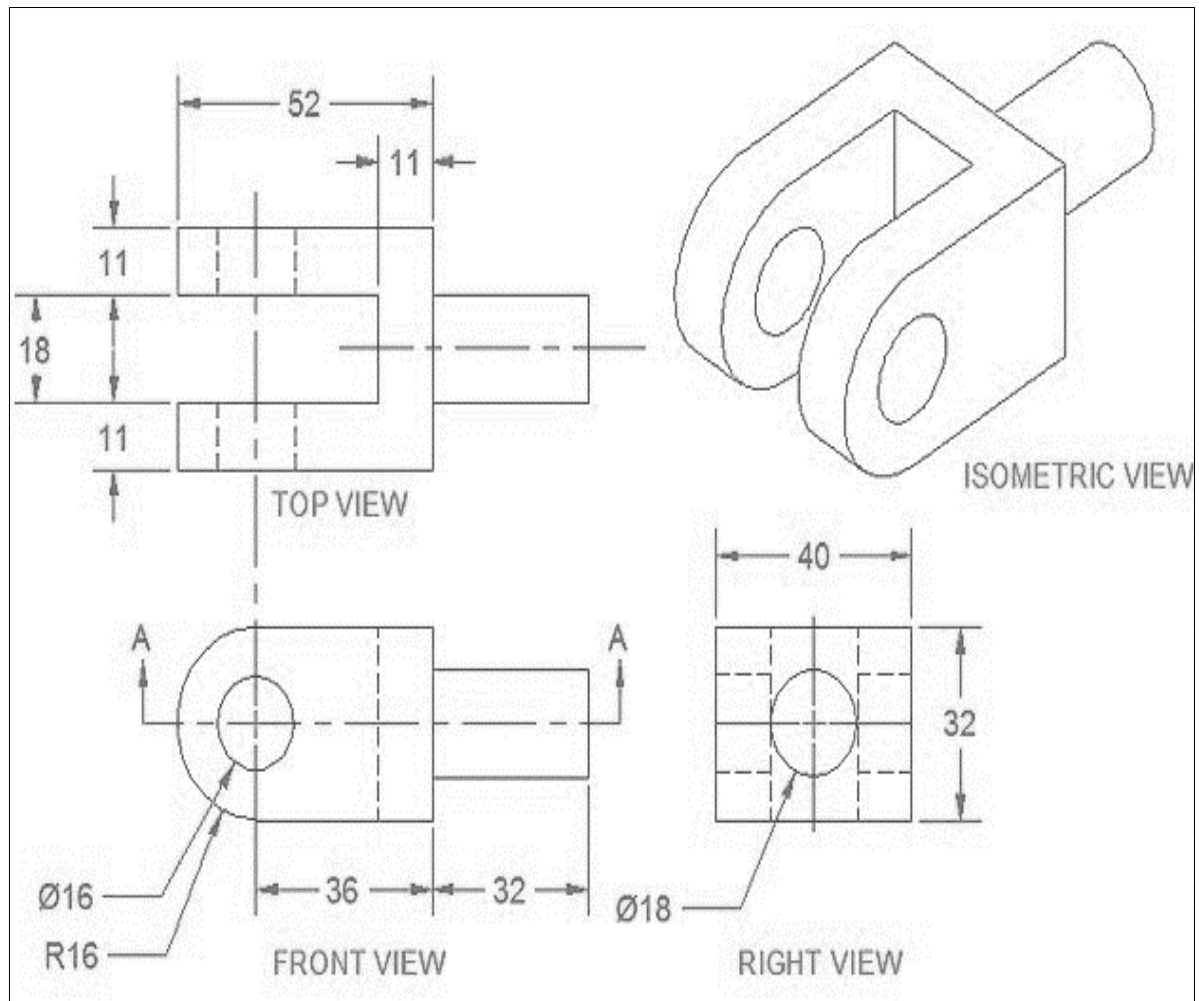
Exercise Number 2: Isometric to Orthographic Views – First angle projection



All Dimensions are in mm

List the Drawing/Modification/Annotation/Dimension tools used for the drawing:

Exercise Number 3: Isometric to Orthographic Views – Third angle projection



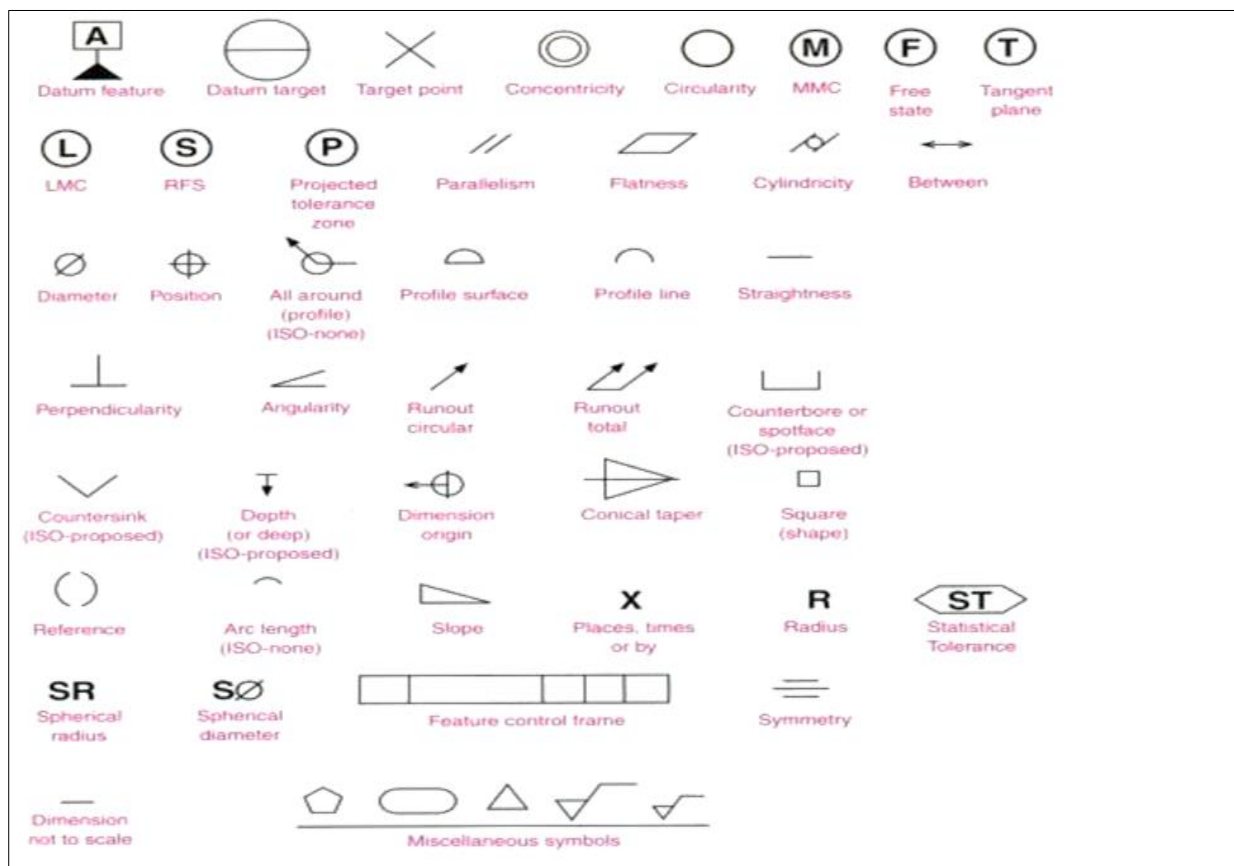
All Dimensions are in mm

List the Drawing/Modification/Annotation/Dimension tools used for the drawing:

Geometric Dimensions and Tolerance

Geometric Category	Tolerance Type	Description	Symbol
For Individual Features	Form	Straightness	
		Flatness	
		Circularity	
Individual or Related Features	Profile	Line Profile	
		Surface Profile	
For Related Features	Orientation	Angularity	
		Perpendicularity	
		Parallelism	
	Location	Position	
		Concentricity	
		Symmetry	
	Runout	Circular Runout *	
		Total Runout *	

Source: <https://images.app.goo.gl/wJS6V44fPdmncwaBA>



Source: <https://bndtechsource.ucoz.com/GDT/Geometric-Dimension-Ing-Tolerancing>

Technical drawing of a mechanical part, showing a cross-section and a top view.

Cross-section View:

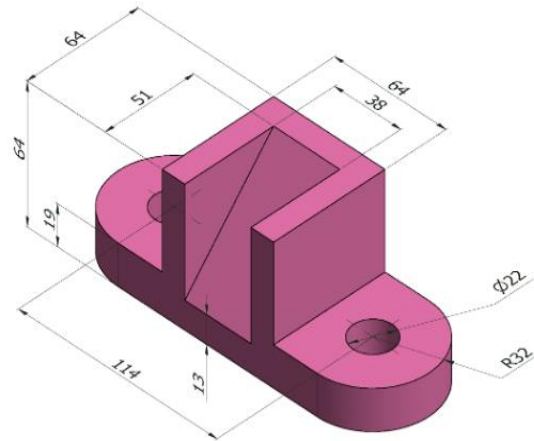
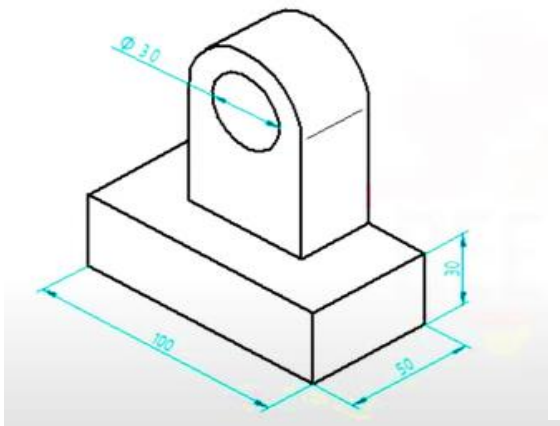
- Dimensions: 9.6, 9.4, 6.6, 6.1, 44.60, 44.45, 36.0, 35.5, 31.8, 31.6, 9.6, 9.4, 25.5, 25.4, 36, 35, 41.3, 41.1.
- Surface texture: $\sqrt{0.06}$ A, $\sqrt{0.08}$ A, $\sqrt{0.14}$ A C, $\sqrt{0.1}$ B, $\sqrt{0.02}$.
- Geometric features: $\text{M42} \times 1.5 - 6g$, $\phi 20.00 - 20.13$, $\phi 0.10$ B, $\phi 0.10$ A C, $\phi 0.10$ B.

Top View:

- Dimensions: 57.6, 56.6, 100, 99, 86.
- Surface texture: $\sqrt{0.14}$ A C, $\sqrt{0.1}$ B.
- Geometric features: 8X $\phi 7.9 - 8.1$, 8X 45° .

Mention the nomenclature of any one feature control frame given in the drawing.

Exercise Number 5: Draw the given isometric views using the ARES commander 3D modelling tool and provide the necessary dimensions.



All Dimensions are in mm

List the Drawing/Modification/Annotation/Dimension tools used for the drawing:
